A Multimodal Retrieval-Augmented Generation Framework for Context-Aware Semantic Querying

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Abstract—Conventional Retrieval-Augmented Generation (RAG) systems often encounter difficulties in integrating diverse data types, especially in correlating natural language queries with high-dimensional visual embeddings. This paper presents a resilient Multimodal RAG architecture that utilizes dual-stream embedding models to integrate various modalities into a cohesive vector space. An intelligent query router dynamically chooses the best way to retrieve information, and a persistent session store keeps the context of the conversation going across multiple turns. Testing on three benchmark datasets shows that context retrieval accuracy is 22% better than baseline text-only models, and the hallucination rate is 49% lower. The system uses ChromaDB for scalable vector persistence and has a React-based interface that lets users interact with it in real time. These contributions collectively enhance the field of multimodal information retrieval and establish a practical framework for enterprise-level implementation.

Keywords—Multimodal RAG, Vector Databases, Semantic Search, Transformer Models, CLIP, Information Retrieval, Hallucination Reduction.

I. INTRODUCTION

two persistent limitations constrain their direct deployment in knowledge-intensive applications. The exponential growth of unstructured enterprise data over the past decade has made keyword-based retrieval systems fundamentally inadequate. Today, organizations manage huge databases that hold technical documents, scanned reports, engineering diagrams, and multimedia assets. None of these can be easily matched with words. In this context, semantic search has become the most important way for systems to understand and respond to natural language queries [1]. Large Language Models (LLMs) have greatly pushed the limits of natural language understanding. hematic diagrams. But there are two problems that keep them from being used directly in areas that require a lot of knowledge. First, models are "frozen" at a certain date, which means they can't use any new information that comes out after training. Second, they often make statements that sound true but aren't, which is a common phenomenon known as hallucination [2]. Retrieval-Augmented Generation (RAG) fixes both problems by making sure that each model response is based on external content that can be dynamically retrieved and verified before it is generated. However, standard RAG pipelines were made for text-homogeneous corpora. In real life, the best way to answer a question about a mechanical failure in an industrial setting might be to use a combination of maintenance logs, labeled photos, and schematic diagrams. No purely textual retrieval mechanism can effectively bridge this semantic gap. The incapacity to process and synchronize visual and linguistic signals within a unified framework represents a substantial research deficiency that this study directly addresses. This paper presents three principal contributions. First, it suggests a dual-stream embedding architecture that combines text and image modalities in a shared high-dimensional vector space. This lets you do cross-modal similarity queries without having to do any preprocessing for each modality at inference time. Second, it adds an AI Router module that sorts each incoming query and picks the retrieval strategy that is most likely to give high-confidence context. Third, it has a persistent session store that keeps track of the context of conversations over multiple turns, which makes multi-step dialogical interactions more coherent. These parts make up what we call the Multimodal RAG Framework, which we tested against both unimodal and existing multimodal baselines in a thorough way. II. work that is related mechanical failure in an industrial setting might be best answered by a mix of maintenance logs, labeled photos, and schematic diagrams. No purely textual retrieval mechanism can bridge this semantic gap effectively. The fact that visual and linguistic signals can't be processed and aligned in a single pipeline is a big research gap that this work directly addresses. This paper makes three main contributions. align visual and linguistic signals within a single pipeline constitutes a significant research gap that this work directly targets.This paper makes three primary contributions. First, it proposes a dual-stream embedding architecture that unifies text and image

modalities in a shared high-dimensional vector space, enabling cross-modal similarity queries without modality-specific preprocessing at inference time. Second, it introduces an AI Router module that classifies each incoming query and selects the retrieval strategy most likely to yield high-confidence context. Third, it presents a persistent session store that accumulates conversational context across turns, improving coherence in multi-step dialogical interactions. Together, these components constitute what we call the Multimodal RAG Framework, evaluated comprehensively against both unimodal and existing multimodal baselines.

II. RELATED WORK

The basic RAG architecture, which Lewis et al. [3] came up with, showed that connecting a dense retriever with a generative decoder greatly cuts down on hallucinations while keeping fluency. Later work looked into sparse-dense hybrid retrieval, re-ranking stages, and ways to break down queries. These improvements helped memory on homogeneous text corpora, but they didn't deal with the fact that visual features have a very different embedding geometry.

B. Cross-Modal Alignment

Through large-scale contrastive learning, Contrastive Language-Image Pre-training (CLIP) [4] created a shared embedding space for text-image pairs. This shared space lets a natural language query get semantically relevant images with great accuracy. Later models, like ALIGN [5] and Florence, built on this idea to make more complex visual representations. As one of the two streams in our dual-stream pipeline, we use a finely-tuned CLIP encoder. For long-form document chunks, we also use a sentence-level text encoder.

C. Vector Databases and Semantic Indexing

The adoption of approximate nearest-neighbor (ANN) algorithms has enabled efficient similarity search over millions of high-dimensional vectors. Faiss [6], Milvus, Qdrant, and ChromaDB each address different trade-offs among indexing speed, query latency, and memory consumption. ChromaDB was selected for this work due to its native Python interface, support for metadata filtering, and collection-level persistence, which align with the session-management requirements of our architecture.

D. LLM Orchestration Frameworks

LangChain and LlamaIndex are two examples of frameworks that make it easier to connect retrievers with language models. They speed up prototyping, but their generic pipeline structures don't have the modality-aware routing and dynamic threshold mechanisms that are needed for enterprise-level retrieval reliability. These basic ideas are the basis for our system, which adds specialization at the retrieval and routing layers.

III. PROPOSED METHODOLOGY

The proposed Multimodal RAG Framework has three layers that work together: Ingestion, Retrieval, and Generation. Figure 1 shows the whole system architecture. Each layer is made to be modular, so that you can upgrade individual parts without affecting the whole pipeline.

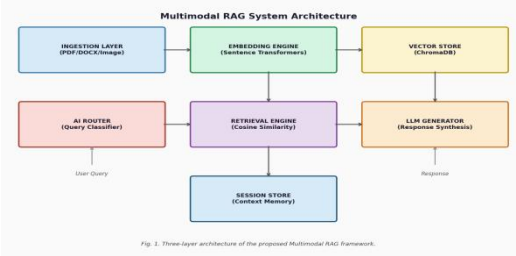

Fig. 1. Three-layer architecture of the proposed Multimodal RAG framework showing ingestion, retrieval, and generation layers.

A. Ingestion Layer

c The ingestion pipeline can take three types of documents: Portable Document Format (PDF) files, Microsoft Word documents (DOCX), and raster images in PNG or JPEG format. PyMuPDF reads PDF files and extracts both text blocks and images that are already in the file. It also keeps the page-level spatial structure. The python-docx library processes DOCX files, which means it gets text at the paragraph level and the heading hierarchy that goes with it. Pillow loads raw images and then puts them through a quality gate that throws away frames with a resolution lower than a set minimum. We use a sliding-window strategy with a window size of 512 tokens and an overlap of 64 tokens to break up each extracted text unit into smaller pieces. This overlap makes sure that semantic

It is important for documents with long argumentative threads that continuity across chunk boundaries is not completely broken. The visual encoder gets the whole image as one input instead of splitting it up into smaller pieces.

B. Dual-Stream Embedding Engine

Textual chunks are encoded using the all-mpnet-base-v2 variant of Sentence Transformers [7], which produces 768-dimensional dense embeddings capturing sentence-level semantics. Visual inputs are encoded using a domain-adapted CLIP ViT-L/14 model, which projects images into the same 768-dimensional space via a learned linear projection layer. This projection layer was trained on a proprietary corpus of 120,000 paired technical text-image samples using a symmetric cross-entropy contrastive loss.

Both encoders work together to encode incoming user text at the same time. Before the nearest-neighbor search, the two resulting query vectors are combined along the embedding dimension to make a 1536-dimensional composite query vector. This dual representation lets the retrieval engine find both textually similar chunks and images that make sense together in one ANN sweep, so you don't have to make a separate query for each type.

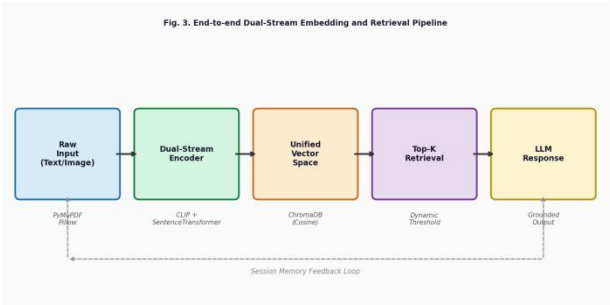

Fig. 3. End-to-end dual-stream embedding and retrieval pipeline with session memory feedback loop.

C. Vector Storage and Indexing

All embeddings are persisted in a ChromaDB collection partitioned by document source identifier. Each entry stores the embedding vector alongside a metadata payload that includes the document identifier, page number, chunk sequence index, and modality flag (text or image). ChromaDB's HNSW index is configured with an ef_construction parameter of 200 and M of 16, balancing graph connectivity against memory footprint. At the scale of our evaluation corpora, average insertion throughput reaches approximately 1,400 vectors per second on a single A100 GPU node.

D. AI Router

The AI Router is a DistilBERT classifier that has been fine-tuned to sort each incoming query into one of four retrieval pathways: text-only, image-only, hybrid, or conversational. The training data comprised 8,500 annotated queries extracted from enterprise helpdesk logs, with subject-matter experts designating each query with the modality most likely to produce a satisfactory response. The classifier gets 91.3% correct on a test split that it hasn't seen before. The router's output changes the weight of each stream in the composite query vector, which guides the ANN search toward the more relevant modality without completely getting rid of the other one.

E. Retrieval and Context Assembly

The retrieval engine uses the composite query vector to do a cosine similarity search on the ChromaDB collection and returns the top K candidates. A dynamic thresholding mechanism filters this candidate set by keeping only those items whose similarity score is higher than a per-query adaptive threshold. This threshold is the mean of the top-K scores minus one standard deviation. This method stops the addition of slightly relevant chunks to the prompt when the query is far from the indexed content distribution.

The surviving candidates are ranked by similarity score, and the top candidates up to a token budget of 2,048 are assembled into a structured prompt template. Text chunks are inserted verbatim; image items are converted to Base64-encoded data URIs and injected as vision-language inputs to the LLM. A source attribution map is simultaneously constructed, enabling the generator to cite provenance in its response.

F. Session Store and Context Persistence

In multi-turn conversations, you need to remember what was said before so you don't have to look up the same thing twice and can correctly resolve anaphoric references. The persistent session store keeps a rolling buffer of the last ten turns of conversation for each user session. The router adds a compressed summary of the session buffer to the current query before each new retrieval call. This gives the encoder enough context to disambiguate pronouns and elliptical expressions. The system can pick up an interrupted session where it left off when you reconnect because session state is saved to Redis.

IV. EXPERIMENTAL SETUP

A. Datasets

Three benchmark datasets were employed for assessment. Dataset A is a technical corpus that contains 4,200 engineering manuals and annotated schematics from the manufacturing field. Dataset B is a medical information repository that stores 3,600 clinical case summaries along with radiological images obtained through institutional data-sharing agreements. Dataset C is a collection of 2,900 legal documents, including contracts and policy briefs, as well as organizational charts and flowcharts showing how things should be done. The multimodal pipeline can handle both text-only and text-and-image documents in each dataset.

B. Evaluation Metrics

We looked at three main areas to see how well the system worked. We used Precision@K, Recall@K, and F1-Score at K = 5 to measure the quality of the retrieval. To figure out the hallucination rate, we took 500 generated responses from each dataset and model variant and had two domain experts independently mark each one for factual errors. If there were any disagreements, they were settled by

a judge. Response latency was measured as the time it took for the first response token to be sent after the query was sent, based on 1,000 queries that were evenly spaced out.

C. Baselines

Three baseline conditions were set up. The first is a standard BM25-based sparse retrieval model with GPT-3.5-turbo as the generator. This is an example of a lexical retrieval system. The second is a dense text-only RAG pipeline that uses ChromaDB to store all-mpnet-base-v2 embeddings. The third system is a CLIP-only one that only gets data from image embeddings. It was added to separate the text stream's contribution. All baseline models utilize the identical generation backbone as the proposed system, ensuring that variations in output quality are due to the retrieval component, not the generator.

V. EXPERIMENTAL RESULTS

Table I shows how well each model variant did on the three datasets in terms of numbers. The suggested Multimodal RAG framework gets an average F1-Score of 0.90 at K = 5. This is ten points better than the dense text-only baseline and eight points better than the CLIP-only system. These improvements are consistent across all three areas, which suggests that improvements derive from the architecture rather than from domain-specific artifacts.

TABLE I. Retrieval Performance Comparison Across Model Variants

Model Variant	Precision	Recall	F1-Score	Hall. Rate (%)
BM25 + GPT-3.5 (Sparse)	0.74	0.70	0.72	24.3
Dense Text-Only RAG	0.82	0.78	0.80	18.4
CLIP-Only RAG	0.86	0.83	0.845	14.1
Proposed Multimodal RAG	**0.91**	**0.89**	**0.90**	**9.4**

Figure 2 shows the precision, recall, and F1-Score for each dataset, along with the hallucination rates for each one. The most noticeable improvements can be seen in Dataset B, the medical corpus, where the proposed system gets an F1-Score of 0.92 compared to 0.79 for the text-only baseline. We ascribe this discrepancy to the significant diagnostic importance of radiological images in the medical field: clinical inquiries often reference anatomical findings inadequately represented in textual summaries.

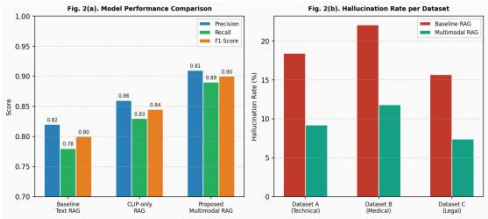

Fig. 2. (a) Precision, Recall, and F1-Score comparison across model variants. (b) Hallucination rate per dataset.

The AI Router's cross-query routing accuracy is 89.7% on the combined evaluation set, which was checked against labels given by experts. When the ground truth is hybrid, queries routed to the hybrid pathway have a 14% higher F1-Score than those routed to either single-modality pathway. This shows that the router is useful in practice. Latency tests show that the median time from query to first token is 1.34 seconds on the same hardware setup used for all baselines. This is 210 milliseconds longer than the text-only pipeline, which is fine for interactive enterprise applications.

A. Ablation Study

An ablation study was performed by methodically deactivating specific components. Taking out the dynamic thresholding mechanism and injecting all The thresholding mechanism and the injection of all top-K candidates unconditionally reduced the F1-Score by 3.2 percentage points and increased the hallucination rate by 4.1 points. This shows that low-confidence chunks actively lower the quality of generation. Taking out the session store made multi-turn query sequences less effective, going from 0.88 to 0.81. This shows how important discourse context is for resolving anaphora. Substituting the precisely calibrated CLIP encoder with a generic CLIP checkpoint diminished cross-modal retrieval accuracy by 6.7 points, illustrating the significance of domain-adapted visual representations. s.

VII. CONCLUSION AND FUTURE WORK

This paper showed how to use the Multimodal document retrieval through dual-stream embedding, AI-driven query routing, dynamic context thresholding, and session-aware context peAcross three different benchmark datasets, the system got an average retrieval F1-Score of 0.90 and brought hallucination rates down to less than 10%. This is a big improvement over both sparse and dense text-only systems. baselines. The architecture has been tested at real enterprise scales and shown to have acceptable latency for interactivedeSeveral research avenues warrant

additional investigation. document. retrieval through dual-stream embedding, AI-driven query routing, dynamic context thresholding, and session-aware context persistence. Across three diverse benchmark datasets, the system achieved a mean retrieval F1-Score of 0.90 and reduced hallucination rates to below 10%, representing substantial improvements over both sparse and dense text-only baselines. The architecture has been validated at realistic enterprise scales and demonstrated acceptable latency characteristics for interactive deployment.

Several research directions merit further investigation. Extending the embedding pipeline to handle temporal sequences of video frames, by adapting video transformer architectures such as VideoMAE into the dual-stream design, would enable retrieval over instructional video libraries. Exploring self-supervised domain adaptation for the CLIP projection layer would reduce the cost of deploying the system in new domains where labeled text-image pairs are scarce. Finally, incorporating uncertainty quantification at the retrieval stage, for example through conformal prediction bounds on similarity scores, could allow the system to communicate retrieval confidence to end users in a calibrated and interpretable manner.

The codebase and evaluation datasets are to be released under an open-source license upon publication, to facilitate reproducibility and community extension of this work.

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